Advanced Joining of High-Performance Metallic Materials

Transient liquid phase bonding of nickel-based superalloys to stainless steel, from processing to properties

ABSTRACT

Are you interested in materials science, physical metallurgy, and advanced microscopy? Join our research team at the University of the Basque Country (UPV/EHU), Spain, and contribute to cutting-edge research on metallic materials for demanding industrial applications.

This PhD project focuses on the development and characterization of dissimilar joints between nickel-based superalloys, such as Inconel 718 and Haynes alloys, and stainless steel, using Transient Liquid Phase Bonding (TLPB). This joining technique offers exciting opportunities to combine the excellent high-temperature performance of superalloys with the cost-effectiveness of stainless steel, particularly for applications in energy generation, aerospace, and corrosive environments.

The research will investigate how processing conditions influence microstructure, mechanical performance, and corrosion resistance. You will gain hands-on experience with advanced characterization techniques, including scanning and transmission electron microscopy (SEM/TEM), electron backscatter diffraction (EBSD), energy-dispersive X-ray spectroscopy (EDS), and focused ion beam (FIB) microscopy, alongside mechanical and electrochemical testing.

The project is supported by an international collaboration with research groups in Austria and Argentina, offering opportunities for international research stays and interdisciplinary teamwork.

Join us to explore the fascinating relationship between processing, microstructure, and properties, and develop the experimental and analytical skills needed for a career in advanced materials research.

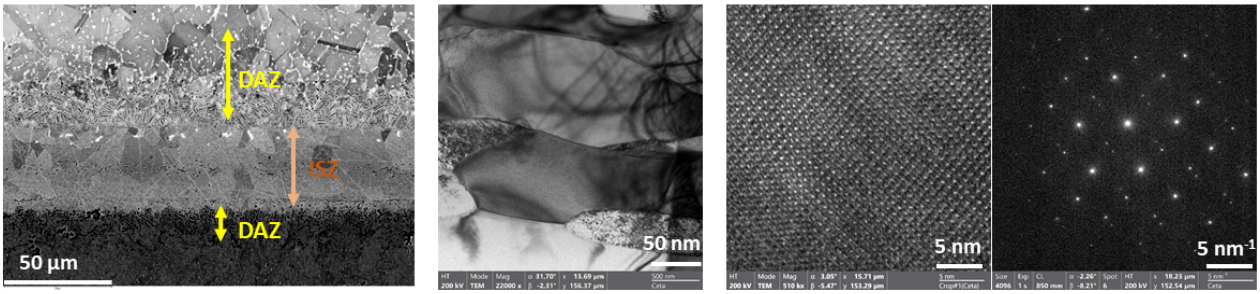


Figure 1. Multiscale characterization of the dissimilar join using scanning and transmission electron microscopy (SEM and TEM), from the micrometre to the nanometre scale. The PhD project will involve advanced microstructural characterization to investigate the relationships between processing conditions, microstructure, and the mechanical and corrosion properties of dissimilar metallic joints.

REQUIREMENTS

Nobody is expected to bring all of this. The degrees are the formal part; the rest is welcome and is taught along the way.

- Bachelor's degree in Physics, Chemistry or Engineering, and a Master's degree in Materials Science and Engineering or a related area, as required for admission to a doctoral programme (or expected completion before the start date).
- Welcome, not required: some background in Materials Science and Engineering or Mechanical Properties of Materials; previous experience in electron microscopy and X-ray diffraction (XRD); some programming, ideally in Python, for data analysis.
- Motivation to pursue experimental research, interest in international research stays (Austria and Argentina), willingness to work in a team, and English good enough to read, write and present scientific work (B2 or higher).

Full or partial funding is available for the thesis projects.

SUPERVISORS

Iñaki López-Ferreño and Gabriel A. López
thermomat research group, Fisika Saila, EHU

CONTACT

Prof. Gabriel A. López
gabrielalejandro.lopez@ehu.eus

this offer online